

Question 1

$$\begin{aligned}x_1 + 2x_2 + 3x_3 &= 0, \\4x_1 + 5x_2 + 6x_3 &= 0, \\7x_1 + 8x_2 &= 0.\end{aligned}$$

We have 3 equations and 3 variables. In matrix form, we have

$$Ax = b, \text{ with } A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

We transform the matrix into a row echelon form to identify its rank

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 0 \end{pmatrix} \stackrel{(1)}{=} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -21 \end{pmatrix} \stackrel{(2)}{=} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 9 \end{pmatrix}$$

where the following elementary row operations have been applied:

$$(1) \quad r_3 = r_3 - 7r_1 \text{ and } r_2 = r_2 - 4r_1.$$

$$(2) \quad r_3 = r_3 - 2r_2.$$

Note that to obtain a true row echelon form, we would also need to normalize each row to obtain a one as the first leading element. However, it is not necessary in order to obtain the rank of the matrix, which is the number of non zero rows in this representation. As each row contains non zero elements, the rank is 3 here.

The system is compatible, as the vector

$$x = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

is a solution of the system. As the matrix is of full rank, there is no redundant equation, and x is the unique solution.

Question 2

$$\begin{aligned}x_1 + 2x_2 + 3x_3 &= 0, \\4x_1 + 5x_2 + 6x_3 &= 0, \\7x_1 + 8x_2 + 9x_3 &= 0.\end{aligned}$$

We have 3 equations and 3 variables. In matrix form, we have

$$Ax = b, \text{ with } A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

We transform the matrix into a row echelon form to identify its rank

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \stackrel{(1)}{=} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & -6 & -12 \end{pmatrix} \stackrel{(2)}{=} \begin{pmatrix} 1 & 2 & 3 \\ 0 & -3 & -6 \\ 0 & 0 & 0 \end{pmatrix}$$

where the following row operations have been applied:

$$(1) \quad r_3 = r_3 - 7r_1 \text{ and } r_2 = r_2 - 4r_1.$$

$$(2) \quad r_3 = r_3 - 2r_2.$$

As only two rows contain non zero elements, the rank of the matrix is 2. The system is compatible, as the vector

$$x = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

is a solution of the system. Therefore, one of the constraints is redundant, and there is an infinite number of solutions.

Question 3

$$x_1 + 2x_2 + 3x_3 = 0,$$

$$2x_1 + 4x_2 + 6x_3 = 0,$$

$$5x_1 + 10x_2 + 15x_3 = 0.$$

We have 3 equations and 3 variables. In matrix form, we have

$$Ax = b, \text{ with } A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 5 & 10 & 15 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

We transform the matrix into a row echelon form to identify its rank

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 5 & 10 & 15 \end{pmatrix} \stackrel{(1)}{=} \begin{pmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

where the following row operations have been applied:

$$(1) \quad r_3 = r_3 - 5r_1 \text{ and } r_2 = r_2 - 2r_1.$$

As only one row contains non zero elements, the rank of the matrix is 1. The system is compatible, as the vector

$$x = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

is a solution of the system. Therefore, two of the constraints are redundant, and there is an infinite number of solutions.

Question 4

$$x_1 + x_2 + x_3 = 1,$$

$$x_1 - x_2 + x_4 = 1,$$

$$x_1 - 5x_2 - 2x_3 + 3x_4 = 1.$$

We have 3 equations and 4 variables. In matrix form, we have

$$Ax = b, \text{ with } A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \\ 1 & -5 & -2 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

We transform the matrix into a row echelon form to identify its rank

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \\ 1 & -5 & -2 & 3 \end{pmatrix} \stackrel{(1)}{=} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & -2 & -1 & 1 \\ 0 & -6 & -3 & 3 \end{pmatrix} \stackrel{(2)}{=} \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & -2 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

where the following row operations have been applied:

$$(1) \quad r_3 = r_3 - r_1 \text{ and } r_2 = r_2 - r_1.$$

$$(2) \quad r_3 = r_3 - 3r_2.$$

As only two rows contain non zero elements, the rank of the matrix is 2. The system is compatible, as the vector

$$x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

is a solution of the system. Therefore, one of the constraints is redundant, and there is an infinite number of solutions.

Question 5

$$x_1 + x_2 = 0,$$

$$x_1 + 2x_2 + 2x_3 = 0,$$

$$3x_2 + 3x_3 = 0.$$

We have 3 equations and 3 variables. In matrix form, we have

$$Ax = b, \text{ with } A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 2 \\ 0 & 3 & 3 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

We transform the matrix into a row echelon form to identify its rank

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 2 & 2 \\ 0 & 3 & 3 \end{pmatrix} \stackrel{(1)}{=} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 3 & 3 \end{pmatrix} \stackrel{(2)}{=} \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 2 \\ 0 & 0 & -3 \end{pmatrix}$$

where the following row operations have been applied:

$$(1) \quad r_2 = r_2 - r_1.$$

$$(2) \quad r_3 = r_3 - 3r_2.$$

As each row contains non zero elements, the rank of the matrix is 3. The system is compatible, as the vector

$$x = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

is a solution of the system. Therefore, no constraint is redundant, and x is a unique solution.